

Po-Yen (Brian) Lu

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EDUCATION

National Tsing Hua University (NTHU)

Hsinchu, Taiwan

M.S. Student in Electrical Engineering

Sep. 2023 - Aug. 2026

- Advised by Prof. Chao-Tsung Huang

B.S. in Electrical Engineering

Sep. 2019 - Jun. 2023

- Selected Courses: Introduction to IC Design, Computer Architecture, Advanced Digital Design Flow
- Overall GPA: 4.2/4.3 (Rank: 4/109)

RESEARCH INTEREST

Algorithm exploration, energy-efficient computing architecture design, advanced FinFET digital design flow, digital VLSI system implementation for deep neural networks, computational photography, and generative AI.

PUBLICATIONS

- **Po-Yen Lu**, Po-Wei Chen, Shau-Hua Yeh, Zhi-Jun Zhang, Yong-Tai Chen, Ting-Yu Wang, Chao-Tsung Huang, “Tiamat: A 98-to-134ms/Step Transformer-Based Diffusion Model Processor Supporting Classifier-Free Guidance for Image Generation” in IEEE International Solid-State Circuits Conference (ISSCC), 2026, accepted.
- Yong-Tai Chen, Yen-Ting Chiu, Yu-Chou Lee, **Po-Yen Lu**, Mu-Hsien Lee, Jia-Syuan Chen, Kuan-Hsien Ho, Hong-Chuan Liao, Chieh-Cheng Chen, Chao-Tsung Huang, “CATTUS: A 4K-UHD 30FPS Deep Image Processor for Channel Attention Equipped U-Net Acceleration in 16nm FinFET” in IEEE Symposium on VLSI Technology and Circuits, 2025.

RESEARCH EXPERIENCE

Vision Circuits and Systems Lab, NTHU

Hsinchu, Taiwan

M.S. Student, Undergraduate Researcher | Advisor: Prof. Chao-Tsung Huang

Feb. 2023 - Present

- Personal High-performance Computing System for Image Generative AI Inference
 - Led a project that developed an energy- and memory-efficient processor for high-quality and real-time image diffusion model inference on edge devices.
 - Proposed a computing architecture optimized for classifier-free guidance (CFG) inference. Devised a memory-efficient inference datapath with microscaling (MX) data format.
 - Tape-out using TSMC 16nm FinFET technology and established a real-time FPGA SoC demonstration platform.
- Deep Learning Accelerator for Image Deblurring
 - Established a complete frontend and backend cell-based digital design flow and hands-on experience in chip tape-out with TSMC 16nm FinFET technology.

- Explored neural network algorithms for single-image super-resolution. Hardware design and implementations for non-linear functions and quantization.

Digital Sound and People Group, NTHU

Hsinchu, Taiwan

Undergraduate Researcher | Advisor: Prof. Yi-Wen Liu

July 2021 - Jun. 2022

- Detection and Analysis of the Hand Movement in Piano Playing with Triboelectric Nanogenerators
 - Developed a smart glove system for hand movement detection in piano performance, in collaboration with Prof. Zong-Hong Lin on triboelectric nanogenerator integration.
 - Built an initial MIDI-based platform for piano accompaniment.

AWARDS & HONORS

- **Scholarship**, Graduate Admission Academic Excellence, NTHU EE Oct. 2023
- **Honorable Mention**, Integrated Circuit Design Contest Aug. 2023
- **Graduated with Distinction**, NTHU Jun. 2023
- **Honorable Mention**, Undergraduate Project Competition, NTHU EE Jun. 2022
- **Scholarship**, Summer Overseas University Exchange Program, NTHU EE Mar. 2022
- **Academic Excellence Award**, NTHU Fall 2020, Spring 2021, Spring 2022

TEACHING EXPERIENCE

IC Design Laboratory, NTHU

Hsinchu, Taiwan

Teaching Assistant (TA)

Sep. 2024 - Jan. 2025

- Assisted students with lab exercises and assignments, in collaboration with eight other TAs.

Introduction to Semiconductor Design, NTHU

Hsinchu, Taiwan

Teaching Assistant

Sep. 2024 - Jan. 2025

- Held office hours and supported students in lecture materials, in collaboration with two other TAs.

Advanced Digital Design Flow, NTHU

Hsinchu, Taiwan

Teaching Assistant

Feb. - Jun. 2024

- Co-authored lab documents on low-power design, signoff verification, and logic equivalence check using TSMC ADFP 16nm FinFET technology.
- Assisted students with lab exercises, assignments, and final projects, in collaboration with four other TAs.

SKILLS

- **Spoken Languages:** Chinese (Native), English (Fluent)
- **Programming Languages:** Python, C/C++, Verilog, SystemVerilog, TCL
- **CAD Tools:** Simulator (Cadence NC-Verilog, Synopsys VCS), Lint (Synopsys SpyGlass), Debugging (Synopsys Verdi), Logic Synthesis (Synopsys Design Compiler), Physical Implementation (Cadence Innovus), Sign-off (Synopsys PrimeTime, Cadence Voltus), Equivalence Check (Cadence Conformal)